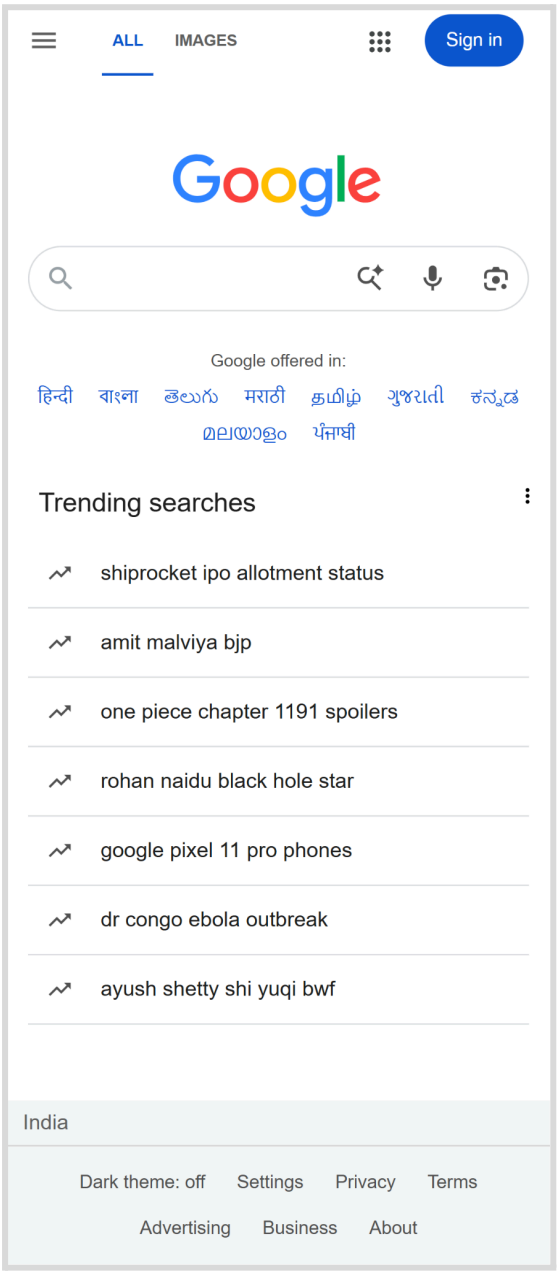
Assignment 1

What happens when I visit the Google homepage from my mobile browser?

Imagine that you have opened the browser application on your mobile phone and typed “www.google.com” on the address bar. You would see something like the following.



Have you thought about what goes behind the scenes for the above content to appear on your browser window? This probably took a few seconds, but an incredible number of operations were done to achieve this.

Your assignment is to try to list down as many steps as possible in this journey.

1. Objectives

At the end of this exercise,

- You should be able to understand how a simple webpage works, and the various technology components working together to make it happen.
- You should be able to get a high level understanding of the server architecture for a typical web application.

2. Assumptions

To simplify the problem statements, please work with the following assumptions.

- You are visiting the webpage for the first time on this particular device. In other words, consider the experience as a non-logged in user.
- You have a good internet connection.
- You are using the Chrome browser.

3. Operations

At a high level, we can group the operations into the following categories.

3.1. Client side operations (Request handling)

The work done by the browser and the operating system to capture the keyboard input and send the request to the network. There may be one or more requests made by the browser to complete the user's request.

3.2. Network operations

There are multiple entities involved in the journey of the request from your phone to the servers at Google.

3.3. Server side operations

The work done by Google's servers to generate the content for the webpage and send it back to the client.

3.4. Client side operations (Response handling)

The work done by the browser and the operating system to render a page similar to the one shown in the screenshot above.

List down as many operations as possible under each of these categories in a Google doc.

4. Optional assignment

(Please note that this part is optional.)
If you are interested in exploring this topic further, repeat the exercise from your laptop or desktop computer.

- Enable the 'Developer tools' option in your Chrome browser. You can enable it by typing Ctlr + Shift + I (on Windows and Linux) and Command + Shift + I (on MacOS), or by clicking ⋮ on the top right and selecting 'More tools' → 'Developer tools'.
- Select the 'Network' tab and observe the behaviour after typing 'www.google.com' on the address bar.
- Are your observations matching those from your experience with the mobile browser?
- What are the different types of requests you see? Are they happening in parallel, or in sequence?
- Do you think 'Developer tools' can be used to improve your understanding of web applications?
- Observe the 'Network' tab after logging into Google. Are the request patterns different from the non-logged in experience?
- What are the additional operations done on the client side and server side for a logged in user?

5. Further reading

https://200ms.thenodebook.com/